



Constructing pie charts

Task A: Introduction to pie charts

1) Izzy collects data on whether there was air frost on 150 days between October and February.

The following table of results is represented in a pie chart.

Each sector of the pie chart represents 6 days worth of data.
How many sectors should there be altogether?

air frost?	frequency
yes	24
no	126

$$150 \div 6 = 25 \text{ sectors}$$

Sunshine over 2 hours?	frequency
yes	135
no	25

2) Aisha collects data on the number of days that had over 2 hours of sun per day on 160 days between April and September. The following table of results is represented in a pie chart.

Aisha draws a pie chart. 27 sectors of her pie chart represent the number of days that did have over 2 hours of sunshine.

How many sectors does Aisha's pie chart have in total?

$$135 \div 27 = 5 \text{ days per sector}$$

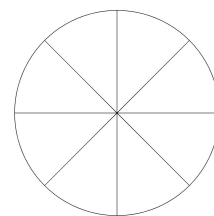
$$160 \text{ total days} \div 5 = 32 \text{ total sectors}$$

3) Alex collects data on the number of days in January (31 days) that had snowfall.

Alex wants to represent this data on a pie chart with 8 equal-sized sectors.

a) Explain why it is impossible for Alex to represent this data in a pie chart with 8 equal-sized sectors.

31 ÷ 8 does not give an integer answer.



snowfall?	frequency
yes	4
no	27

b) Explain whether it is possible for Alex to represent this data in a pie chart with any number of equal-sized sectors.

It is only possible if the pie chart is split into a multiple of 31 sectors.



Task B: Calculating angles for pie charts

1) Complete the calculations to represent these frequencies as angles on a pie chart.

$$360^\circ \div 72 = 5^\circ$$

weather	frequency	angles
sun	37	185
cloud	22	110
rain	13	65
total	72	360°

$\times 5$

2) Complete the calculations to represent these frequencies as angles on a pie chart.

$$360^\circ \div \underline{60} = \underline{6}$$

meal	cost	angles
breakfast	12	72
lunch	23	138
dinner	25	150
total	60	360°

$\times \underline{6}$

3) Complete the calculations to represent these frequencies as angles on a pie chart.

Hint: The multiplier is a non-integer.

weather	hours	
sun	40	180
cloud	24	108
rain	16	72
total	80	360°

$\times 4.5 = \frac{9}{2}$



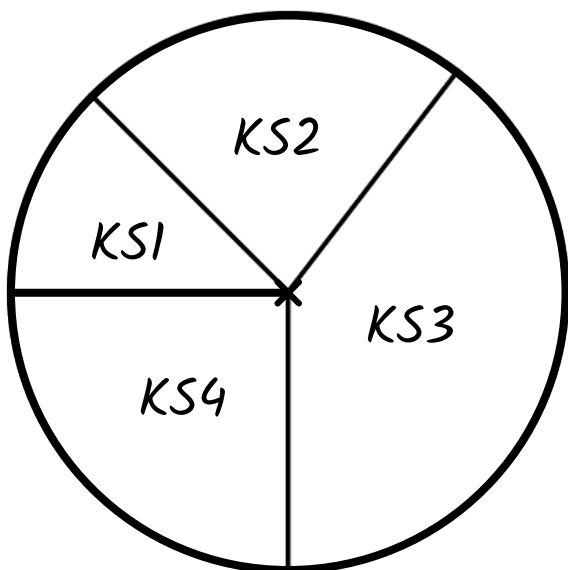
Task C: Constructing pie charts

1) The table shows both the frequency and the angles required to represent the data on a pie chart.

The pie chart has been fully constructed, but not labelled.

Label each sector of the pie chart with the correct Key Stage.

Key Stage	Frequency	Angle
KS1	72	45°
KS2	132	82.5°
KS3	228	142.5°
KS4	144	90°
total	576	360°



3) The table shows both the frequency and most of the angles required to represent the data on a pie chart.

Construct this pie chart.

Footwear	Frequency	Angle
Trainers	42	105°
Boots	68	170°
Flip-Flops	10	25°
Other	24	60°
	144	360°

2) The table shows both the angles required to represent this data on a pie chart.

Key Stage	Angle
Primary	180°
Secondary	75°
College	105°
total	360°

Construct this pie chart.

